Accessible Skimming: Faster Screen Reading of Web Pages

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ABSTRACT
Sighted people know how to quickly glance over the headlines and news articles online to get the gist of information. On the other hand, people who are blind use screen-readers to listen through the content narrated by a serial audio interface. This interface does not give them an opportunity to know what content to skip and what to listen to. In this work, I present an automated approach to facilitate *non-visual skimming* of web pages.

Categories and Subject Descriptors
H.5.2 [**Information Interfaces and Presentation**]: User Interfaces; H.5.4 [Information Interfaces and Presentation]: Hypertext/Hypermedia – *navigation.*

General Terms
Human Factors; Experimentation; Design.

Keywords
Skimming; Assistive Technology; Blind; User Interface; Screen Reader; Web Browser; Audio Interface; Accessibility.

1. INTRODUCTION
In our information-driven web-based society, we are all gradually falling "victims" to *information overload* [2]. However, while sighted people are finding ways to sift through information faster, Internet users who are blind are experiencing an even greater information overload. These people access computers and Internet using screen-reader software, which reads the information on a computer screen sequentially using computer-generated speech. So, they either listen to all of the content or listen to the first part of each sentence or paragraph before they skip to the next one. In this work, I design a novel skimming algorithm, outline a non-visual skimming interface, and report on our user study with 23 screen-reader users. The results demonstrate that the resulting algorithm for automatic summarization can be successfully used for non-visual skimming.

2. BACKGROUND AND RELATED WORK
When sighted users look at a random web page, their quick eye movements, called saccades, help them get a quick overview of the page and find what they need. The structure and formatting of the page further helps them find the information relevant to their goals. In most web pages, this still leaves them a lot of content to read through, imposing a *heavy cognitive load* [6]. But again, saccades help speed up reading and reduce the cognitive load. The process of quickly glancing over the textual content with the goal of picking up the gist of the content is called *Skimming*.

Screen readers [4, 5, 7] are assistive technology software that narrate the content of the screen using text-to-speech and allow users to navigate through the content marked up by HTML tags for headings, paragraphs, links, buttons, etc. Because of the low bandwidth of a serial audio interface and the fact that one has to hear the information before deciding if it is relevant, people who are blind spend considerably more time identifying the information they need [3]. A little has been done to develop an effective method for skimming, so that blind people could process information faster.

The JAWS screen-reader has a naive implementation of "skimming" [4] that allows users to read the first line or the first sentence of each paragraph. This methodology still does not offer a truly functional tool for skimming, and does not follow the same skimming process employed by sighted people. Our earlier experiment with 20 people [1] reported the usability of non-visual skimming in listening-and-comprehension and in search scenarios. To simulate skimming, we used several variations of human-generated summaries. These experiments demonstrated that skimming enabled people who are blind to read faster while maintaining high level of comprehension and retention.

Visual skimming is akin to the extractive summarization, because when people scan the text they take it as it is. Interviews with 20 screen-reader users [1] also suggested that the extractive summarization approach that preserves the original content is the most suitable technique for enabling non-visual skimming.

3. OVERVIEW OF OUR APPROACH

3.1 Skimming Algorithm
At a high level, our summarization algorithm works as follows.

Firstly, every sentence is parsed to extract grammatical relations amongst its words. Secondly, a lexical tree based on these relations is constructed, where each node of the tree represents a word in the sentence. Thirdly, for every word in this tree, its grammatical (i.e., POS tags) as well as structural features (related to in-degree/out-degree, etc.) are extracted. These features are given to a trained classifier to determine whether or not to include the word in the skimming summary. Finally, a subtree consisting of the selected words is constructed. This subtree represents the summary that users interact with via a non-visual interface.

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3.2 Skimming Interface

Summarization by itself does not satisfy the need for skimming; for screen-reader users, reading a summary is more like reading a separate, albeit shorter, narrative. Therefore, we designed an interface that allowed screen-reader users to switch seamlessly between reading the skimming summary and the original web page preserving the current reading position. This feature helps emulate the behavior of experienced sighted readers who usually can scan quickly and then, at any point, slow down to read the text regularly.

The skimming interface was integrated into our Non-Visual screen-reading platform, which is a Java application that enables a screen reading interface for the Firefox web browser. It supports all the typical navigation shortcuts available in the mainstream screen-readers such as JAWS [4], e.g., paragraph / sentence / word / character navigation, pause / resume, etc.

A shortcut (e.g., Shift+S) lets the user toggle between skimming mode and full text mode. The skimming interface can be invoked on any web page content regardless of its length as long as the content is organized in sentences (e.g., snippets of news articles along with headlines), but lists of links that can be found in menus, on the other hand, should be read in their entirety.

4. USER STUDY

The Disability Resource Center at the Arizona State University helped us recruit 23 participants (7 male and 16 female) who were blind and well-versed in the use of screen readers, with JAWS as their primary screen reader. To evaluate the interface and the underlying algorithm, the participants were asked to perform three tasks in two different scenarios: listening-and-comprehension and searching. For each of the three tasks the participants were using one of the three different kinds of content: Gold-standard (human-generated) summary (G), Skimming summary (S), or the original text of the articles (F).

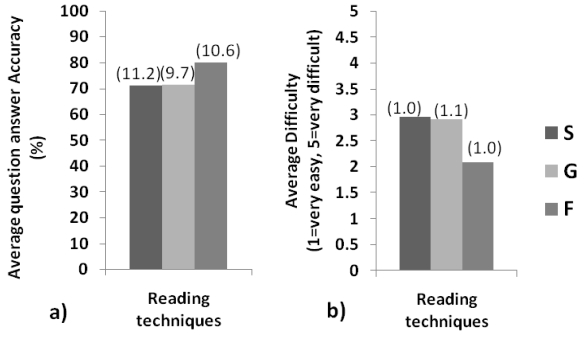


Figure 1. a) Average accuracy and b) Average difficulty (St. Dev.) of question answering with S, G and F

5. HYPOTHESES TESTING

We formulated a series of specific hypotheses to analyze the results of the study. The major hypotheses accepted based on the statistical significance tests are presented below:

H1: After listening to summaries S and G, subjects can answer reading-and-comprehension questions equally well (Figure 1a).

H2: Subjects perceive question-answering tasks to be of the same difficulty after listening to G and F or S and F Figure 1b).

H3: Subjects find information faster while skimming with G/S, than when using screen-reader navigation shortcuts (Figure 2a).

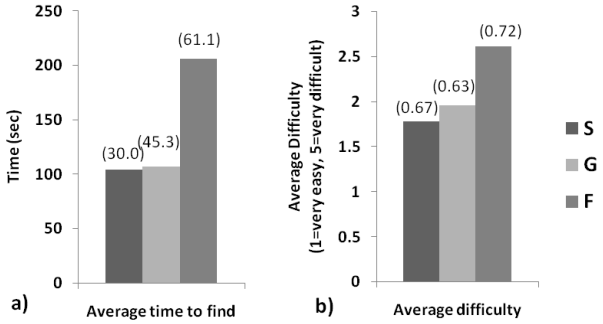


Figure 2. a) Average time and b) Average difficulty in the searching task using skimming vs. regular shortcuts

H4: Subjects perceived ad-hoc searching to be more difficult compared to using skimming S or G (Figure 2b).

H5: Subjects were not able to distinguish between human- and computer-generated summaries.

We observe no difference in performance on these tasks between groups of subjects: **H8:** who lost eyesight before age 20 and those who lost later, **H9:** belonging to different age categories, **H10:** with different comfort level with computers.

6. CONCLUSION AND FUTURE WORK

The approach presented in this work can summarize one sentence at a time without the dependence on the entire document. It can extract meaningful word combinations from most sentences and preserve the original order of the extracted text. All these properties make this summarization approach more suitable for supporting a non-visual skimming interface. Our experiments provide evidence to the claim that non-visual skimming can enable screen-reader users to browse textual content faster.

This work opens the possibilities to further accelerate non-visual skimming by varying the compression ratio of summarization. Skimming can also be extended to touch interfaces. Armed with the ability to change the amount of information as necessary, people who are blind will be able to battle the information overload and cognitive load more effectively, while enjoying the speed-reading commonly used by sighted readers.

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